

Chapter 9 Assignment — Vectors in Two and Three Dimensions

Name: _____

Date: _____

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example*. Then try the *Practice* (skill) and the *Challenge* (deeper) items. Give exact answers when possible. Use unit vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and component notation $\langle a, b, c \rangle$ as appropriate. State angles to the nearest degree if not exact.

1. Topic 1: Vectors in Two Dimensions

Strategy Notes. A vector $\mathbf{u}$ in the plane can be represented as $\langle u_1, u_2 \rangle = \langle x_Q - x_P, y_Q - y_P \rangle$ if it goes from P to Q . Its magnitude is $\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2}$, its direction angle θ satisfies $\tan \theta = \frac{u_2}{u_1}$ with a quadrant check, and the unit vector in its direction is $\frac{\mathbf{u}}{\|\mathbf{u}\|}$.

Worked Example. From $P(3, -1)$ to $Q(-2, 4)$, $\mathbf{u} = \langle -5, 5 \rangle = -5\mathbf{i} + 5\mathbf{j}$. Then $\|\mathbf{u}\| = \sqrt{(-5)^2 + 5^2} = 5\sqrt{2}$ and the direction angle is $\theta = \frac{3\pi}{4}$ (Quadrant II). A unit vector is $\frac{1}{5\sqrt{2}}\langle -5, 5 \rangle = \langle -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \rangle$.

Practice.

- (a) The vector $\mathbf{v}$ has initial point $A(-4, 6)$ and terminal point $B(3, 2)$. Express $\mathbf{v}$ in $\mathbf{i}, \mathbf{j}$ form, find $\|\mathbf{v}\|$, and give its direction angle.

- (b) Find a unit vector in the direction of $\mathbf{w} = \langle 8, -7 \rangle$ and write $\mathbf{w}$ as $a\mathbf{i} + b\mathbf{j}$.

Challenge.

- (a) Nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^2$ are perpendicular. If $\|\mathbf{u}\| = 5$ and $\|\mathbf{u} + \mathbf{v}\| = 13$, find $\|\mathbf{v}\|$.

Strategy Notes. For vectors $\mathbf{u}$ and $\mathbf{v}$, $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2$ (and similarly in $\mathbb{R}^3$). The angle between them satisfies $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$. The projection of $\mathbf{u}$ onto $\mathbf{v}$ is $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}$ and the scalar component is $\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}$.

Worked Example. Let $\mathbf{u} = \langle 3, -1 \rangle$ and $\mathbf{v} = \langle 4, 2 \rangle$. Then $\mathbf{u} \cdot \mathbf{v} = 3 \cdot 4 + (-1) \cdot 2 = 10$. With $\|\mathbf{u}\| = \sqrt{10}$ and $\|\mathbf{v}\| = \sqrt{20}$, we get $\cos \theta = \frac{10}{\sqrt{10}\sqrt{20}} = \frac{1}{\sqrt{2}}$, so $\theta = 45^\circ$. Moreover, $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{10}{20} \mathbf{v} = \frac{1}{2} \langle 4, 2 \rangle = \langle 2, 1 \rangle$.

(a) Compute $\mathbf{u} \cdot \mathbf{v}$ and the angle between $\mathbf{u} = \langle 4, -2 \rangle$ and $\mathbf{v} = \langle 1, 3 \rangle$ (nearest degree).

- (b) Find the scalar component of $\mathbf{u} = \langle 6, 8 \rangle$ along $\mathbf{v} = \langle 4, -3 \rangle$ and determine whether $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.

3. Topic 3: Three-Dimensional Coordinate Geometry

Strategy Notes. For points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, the distance is $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$. A sphere of center (h, k, l) and radius r has equation $(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$.

Worked Example. Center $P(2, -1, 3)$ passing through $Q(6, 1, 3)$ gives $r = \sqrt{(4)^2 + (2)^2 + 0^2} = \sqrt{20}$. Thus the sphere is $(x - 2)^2 + (y + 1)^2 + (z - 3)^2 = 20$.

Practice.

(a) Find the distance between $A(4, 3, -1)$ and $B(6, -1, 3)$ and the midpoint of $\overline{AB}$.

(b) Find the equation of the sphere for which $\overline{CD}$ is a diameter if $C(1, -2, 4)$ and $D(5, 0, -2)$.

4. **Topic 4: Vectors in Three Dimensions**

Strategy Notes. Write $\mathbf{v} = \langle v_1, v_2, v_3 \rangle = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k}$. Magnitude $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$. A unit vector is $\frac{\mathbf{v}}{\|\mathbf{v}\|}$. If the direction cosines are $(\cos \alpha, \cos \beta, \cos \gamma)$ then $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ and $\mathbf{v} = \|\mathbf{v}\| \langle \cos \alpha, \cos \beta, \cos \gamma \rangle$.

Worked Example. From $P(4, 3, -1)$ to $Q(6, -1, 3)$, $\mathbf{v} = \langle 2, -4, 4 \rangle = 2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$. Then $\|\mathbf{v}\| = \sqrt{4 + 16 + 16} = 6$ and a unit vector is $\langle \frac{1}{3}, -\frac{2}{3}, \frac{2}{3} \rangle$.

Practice.

(a) For $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, find $\|\mathbf{v}\|$ and a unit vector in the direction of $\mathbf{v}$.

(b) Find a vector of magnitude 10 whose direction cosines are $(\frac{3}{5}, \frac{4}{5}, 0)$.

5. Topic 5: The Cross Product and Areas

Strategy Notes. For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$, $\mathbf{u} \times \mathbf{v}$ is perpendicular to both with magnitude $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$. Use the determinant formula with $\mathbf{i}, \mathbf{j}, \mathbf{k}$. The area of the parallelogram spanned by $\mathbf{u}, \mathbf{v}$ is $\|\mathbf{u} \times \mathbf{v}\|$; for the triangle it is $\frac{1}{2}\|\mathbf{u} \times \mathbf{v}\|$.

Worked Example. With $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle 1, 4, -2 \rangle$,

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 3 \\ 1 & 4 & -2 \end{vmatrix} = \langle -10, 7, 9 \rangle.$$

Hence $\|\mathbf{u} \times \mathbf{v}\| = \sqrt{230}$ and the triangle area is $\frac{1}{2}\sqrt{230}$.

Practice.

- (a) Compute $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \langle 3, 4, 1 \rangle$ and $\mathbf{b} = \langle -2, 5, 2 \rangle$, and interpret $\|\mathbf{a} \times \mathbf{b}\|$ as an area.

- (b) Find a unit vector orthogonal to both $\mathbf{a} = \langle 1, 0, 2 \rangle$ and $\mathbf{b} = \langle 2, -1, 1 \rangle$.

6. Topic 6: Equations of Lines and Planes

Strategy Notes. A line through $P(x_0, y_0, z_0)$ with direction $\mathbf{v} = \langle a, b, c \rangle$ has vector form $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ and parametric equations $x = x_0 + at$, $y = y_0 + bt$, $z = z_0 + ct$. A plane with normal $\mathbf{n} = \langle A, B, C \rangle$ through P_0 satisfies $\mathbf{n} \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$ (often written $Ax + By + Cz = D$).

Worked Example. Through $P(1, -1, 2)$, $Q(3, 0, 5)$, and $R(2, 1, 1)$, take $\mathbf{u} = \overrightarrow{PQ} = \langle 2, 1, 3 \rangle$ and $\mathbf{w} = \overrightarrow{PR} = \langle 1, 2, -1 \rangle$. Then $\mathbf{n} = \mathbf{u} \times \mathbf{w} = \langle -7, 5, 3 \rangle$, so one plane equation is $-7(x - 1) + 5(y + 1) + 3(z - 2) = 0$, i.e., $7x - 5y - 3z - 6 = 0$.

Practice.

- (a) Find symmetric and parametric equations for the line through $(2, 3, -2)$ parallel to $\langle -1, 4, 2 \rangle$.

- (b) Find an equation of the plane through $A(1, 2, -1)$, $B(3, -1, 2)$, and $C(2, 0, 3)$.

7. Topic 7: Focus on Modeling — Velocity Addition and Bearings

Strategy Notes. Represent speeds and headings as vectors. Take the x -axis as East and the y -axis as North. A heading “N α° E” has components $\langle v \sin \alpha, v \cos \alpha \rangle$. Ground velocity equals the vector sum of the vehicle’s velocity relative to the medium and the medium’s velocity.

Worked Example. A river flows west at 6 mph. A motorboat heads in still water at 15 mph with bearing N 40° E. The velocity relative to water is $\langle 15 \sin 40^\circ, 15 \cos 40^\circ \rangle \approx \langle 9.64, 11.49 \rangle$. Adding the current $\langle -6, 0 \rangle$ gives ground velocity $\langle 3.64, 11.49 \rangle$. The true speed is $\sqrt{3.64^2 + 11.49^2} \approx 12.1$ mph and the course is about $\arctan\left(\frac{11.49}{3.64}\right) \approx 72^\circ$ north of east (bearing E 72° N).

Practice.

- (a) A plane’s airspeed is 200 mph with heading N 60° E. A wind blows 40 mph due east. Find the ground speed and bearing of the plane.
- (b) The pilot wants a due-north ground track while the 40 mph wind blows east. What heading (nearest degree) should the pilot fly, and what is the resulting ground speed?